LEO Satellite Communication Simulation Framework for Connected Vehicles

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***Abstract*—The growing demand of connected vehicles for high-speed, pervasive communication worldwide has led to a surge in the interest of adopting large Low Earth Orbit (LEO) satellite communications. However, the feasibility study and performance evaluation of LEO satellite communication for connected vehicles is challenging due to the high infrastructure costs involved. In this paper, we propose an integrated simulation framework with satellite mobility, vehicle mobility, and network simulators to study and evaluate the LEO satellite communication for connected vehicles. We first create a synthetic LEO satellite constellation with a demonstration of the orbital mobility and preliminary RTT latency over Japan. Then, we further investigate three scenarios in our developed simulation framework, showing that a comparable performance can be achieved refer to the existing commercialized StarLink services.**

***Index Terms*—LEO satellite, simulation, connected vehicles**

I. Introduction

By connecting to the Cloud, connected vehicles offer drivers and passengers a wide range of advanced mobility services including AR/VR-empowered infotainment, predictive emergency calls, and various driving assistance beyond the traditional vehicle experience. To ensure a more personalized and seamless driving experience, these connected services require constant network connectivity. Currently, vehicles connect to the Cloud through cellular or Wi-Fi networks. However, these terrestrial networks always have limited coverage due to insufficient infrastructure or economic reasons. In this sense, satellite communication is considered one of the most promising technologies to meet the needs of connected vehicles for seamless coverage without being affected by terrestrial limitations.

In recent years, there has been a significant rise in competition amongst companies to develop space technology that can provide high-speed Internet connectivity globally at affordable cost by the masses. Among these companies, SpaceX is leading the way by deploying a low-earth orbit (LEO) satellite constellation, which is revolutionizing Internet technology [1]. Thanks to a cost reduction of approximately 1/200 compared to previous methods [2], [3], the deployment of LEO satellite constellations has become technically and economically feasible. The use of LEO satellite communications in connected vehicles has been gaining traction, but the evaluation of service performance is costly and difficult in many areas. To evaluate end-to-end performance from the connected vehicles to the Cloud via LEO satellites and their corresponding ground infrastructure, large-scale experiments are required, which is not always practical. Therefore, simulators are a suitable alternative for evaluating LEO satellite communications for connected vehicles. This study proposes and develops an integrated simulator to evaluate LEO satellite communications for connected vehicles. Additionally, a synthetic LEO satellite constellation is demonstrated for the orbital mobility and preliminary RTT latency over Japan. And three scenarios are shown with our developed simulation framework that a comparable performance can be achieved referring to the existing commercialized StarLink services.

II. Related Works

Numerous communication network simulators related to connected vehicles have been developed in previous studies, as seen in references such as [4]–[7]. The OMNeT++ open-source simulation framework [12] has been widely utilized for network performance evaluations, particularly for vehicular networking. For instance, the study [4] utilized OMNeT++ to construct a time-sensitive network and predict its effectiveness. On the other hand, the work [8]–[10] evaluated the use of LEO satellite communications in IoT and broadband communication services using simulation. In the work [8], the Open Source Satellite Simulator (OS3) framework was developed to simulate satellite constellations while considering factors such as satellite orbit and moving speed. Literature [9] focused on using LEO satellite communications in IoT services and conducted predictive simulations.

The previous studies focused on developing system-level simulators and predicting their effectiveness, with the aim of reducing costs associated with development and experimentation. However, none of them are sufficient for satellite communication in connected vehicles. In this study, we explore simulators designed specifically for examining the use of LEO satellite communication in connected vehicles. Evaluating the simulation of LEO satellite communication for connected vehicles requires the integration of three components: mobile vehicles, satellite constellations, and communication networks. To accurately evaluate the performance of satellite communication for connected vehicles, it is necessary to conduct system simulation evaluations by incorporating vehicles and satellite mobility and their interaction and effects on communications.

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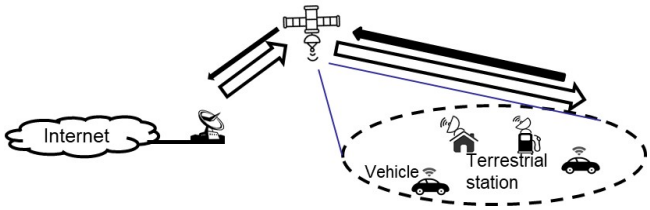

Fig. 1. Considered use case scenario

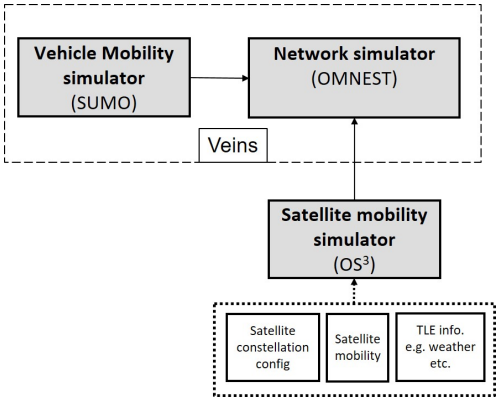

Fig. 2. Architecture of the proposed simulation framework

III. Simulation Framework

A. Architecture of Simulation Framework

The LEO satellite communication network architecture intended for use in connected vehicles is shown in Figure1. The system assumes that mobile terminals, i.e. connected vehicles on the ground, are connected to the Internet via LEO satellites and Earth ground stations. The wireless link between the user terminal and the LEO satellite is generally defined as the service link, while the wireless link between the LEO satellite and the earth station is defined as the feeder link. In the considered use case scenario for connected vehicles, the integrated simulation framework, as depicted in Figure 2, is consisted of SUMO [11] for simulating the movement of vehicles, OS3 for satellites constellation, and OMNEST (commercial version of OMNeT++) for the communication simulation between network nodes. In our proposed integrated simulation framework, we utilized OMNEST [13] as the base simulation modeling tool. It is widely used for simulation modeling and has been used in various previous studies as well.

We illustrate the architecture of the developed simulator in Fig. 2. To simulate the communication network, vehicle mobility, and satellite constellation in an integrated manner, we integrated and co-simulated the widely used simulators such as OMNEST, SUMO, and Open Source Satellite Simulator (OS^3). The communication simulator was built on OMNEST using the Veins (Vehicular Network Simulation) framework [14], which provides a library of communication network simulation for vehicular networking. By connecting Veins with SUMO (Simulation of Urban Mobility), which allows

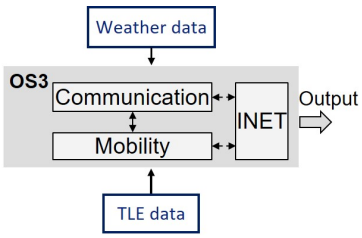

Fig. 3. Architecture of LEO satellite mobility module

the simulation of large-scale traffic networks including urban traffic and transportation systems, we were able to more accurately evaluate the use of connected vehicles under various traffic scenarios.

Furthermore, we utilized the OS^3 framework to simulate the satellite constellation, including the satellite's position based on its orbit, dynamics, weather, and other physical elements. The simulated satellite positions were then fed into the OMNEST network simulator to enable integrated simulation evaluation.

In OMNEST, modules are represented by a hierarchy written in C++. The relationships between modules and their communication links are saved as Network Description (NED) files, which can be graphically modeled. In this study, we utilized communication protocol modules in the INET communication framework to build communication functionalities for network terminals such as vehicles, satellites, and ground stations. Although OMNEST has mobility support, it is limited to simple mobility patterns. Considering the integration of network simulation, it is necessary to simulate wireless communication between nodes by accurately determining the position of connected vehicles and using wireless communication modules.

Therefore, we utilized the microscopic road traffic simulation SUMO, which is based on detailed traffic and traffic scenario integrated into Veins. Furthermore, for LEO satellite constellations, we provided the initial position, velocity, mass, size, and other physical characteristics of the satellite using the OS^3 framework, and calculated the satellite's position using the mathematical model SGP4 (Simplified General Perturbations 4). This resulted in a configuration where the information on the position, velocity, and attenuation of the satellite calculated by OS^3 was passed to OMNEST. However, the coordinate systems in SUMO and OS^3 are different, and the position information of vehicles and satellites can not be used directly in OMNEST. As a result, we adopted the geographic coordinate system to unify the locations of vehicles and satellites with latitude, longitude, and altitude.

B. LEO Satellite Constellation Mobility Module

The implementation of the LEO satellite mobility module is based on the OS^3 framework and uses the mathematical model SGP4 (Simplified General Perturbations 4) to calculate the satellite's position, taking into account elements such as gravity, atmospheric drag, and solar radiation pressure. The

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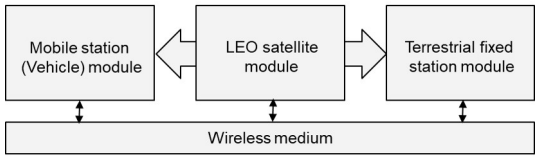

Fig. 4. Architecture of satellite communication module

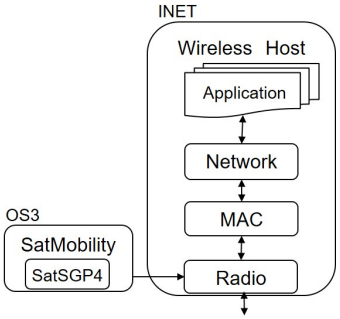

Fig. 5. LEO satellite module

architecture of the LEO satellite mobility module is shown in Figure 3,

To facilitate LEO satellite mobility simulation for a synthetic constellation, a new orbit calculation module was developed, consisting of a sub-module based on Kepler's laws and an orbit calculation sub-module constructed from the two-line element set (TLE) which is a data format encoding a list of orbital elements of an Earth-orbiting object for a given time. The TLE data is specific to model SGP4, is widely used to describe the trajectories of Earth-orbiting objects, including satellites.

C. Satellite Communication Module

The OMNEST communication module was developed to facilitate communication between vehicles, satellites, and ground stations on the network. The communication module architecture is illustrated in Figure 4. The module was developed using the INET library as the basis, with wireless media, physical layer, and other wireless communication functionalities implemented.

Figure 5 depicts the configuration of the LEO satellite module, which is assembled from the Wireless Host and SatMobility submodules. The SatMobility submodule performs precise satellite position simulation by calculating the satellite position and velocity using the previously mentioned OS^3 framework. The Mobility module in OS^3 maps only the longitude and latitude (ignoring altitude) to X, Y, and Z coordinates in OMNEST, but to achieve more accurate distance calculations for determining propagation delay, a new INET-based SatMobility submodule was implemented. Information on the satellite's position is passed from the SatMobility module to the Wireless Host submodule, which performs wireless communication functions. The Wireless Host submodule consists of Radio, Media Access Control (MAC), Network, and Application components. The Radio

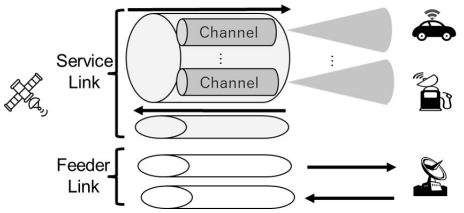

Fig. 6. MAC sub-module of LEO satellite module

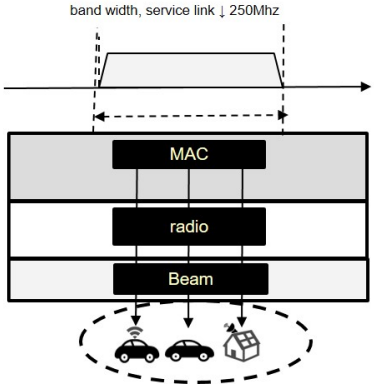

Fig. 7. The multi-access implementation in a satellite beam

component is a wireless transceiver module that performs the actual transmission and reception of wireless signals. Based on information such as the satellite's position from the SatMobility module, the Wireless Host submodule determines the communication distance and radio propagation characteristics. In this context, we considered various attenuation factors, such as path loss due to distance, shadowing fading, attenuation due to environmental factors such as rain, scintillation, and cluster loss, using the 3GPP recommended channel model [15]. The Application layer generates packets for electronic mail, voice, and other applications, and determines the path for packet transmission between the source and destination through the Network layer before passing it to the Media Access Control (MAC). The MAC layer manages wireless media access and passes the physical layer (i.e., the aforementioned Radio module) for transmission when allowed by the protocol (e.g., TDMA).

The multi-beam technology is widely adopted in current satellite systems to improve coverage, capacity, and spectral efficiency. This technology provides communication connections to a large number of user terminals through multi-access control in each beam. In this study, a satellite with multi-beam and multi-access functions was implemented, as shown in Figure 6. The satellite communicates with user terminals on the ground and earth stations through service links (shown in blue) and feeder links (shown in green). These links use different frequencies defined by the International Telecommunication Union (ITU) radio communication regulations for both uplink and downlink communications. In the service link, multi-beams are implemented according to the frequency assigned to each beam, as indicated in Figure 6. Additionally, a MAC and

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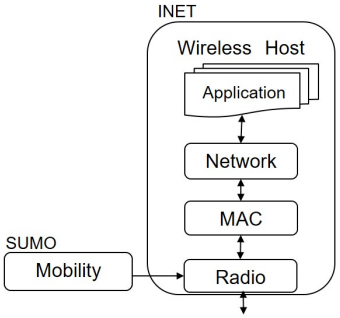

Fig. 8. Mobile station (vehicle) module

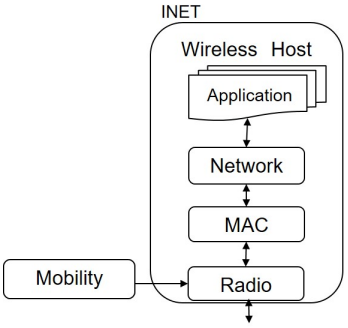

Fig. 9. Terrestrial fixed station module

Radio component were implemented independently in each beam, as shown in Figure 7. To enable multi-access control for multiple user terminals, both Time Division Multiple Access (TDMA) and Frequency Division Multiple Access (FDMA) were implemented.

The vehicle was implemented as a mobile station (vehicle) module, as shown in Figure 8. The Mobility of the mobile station (vehicle) module simulated the vehicle's movement by importing the City Map of Minato Mirai in Yokohama into SUMO. The information on the vehicle's position and velocity obtained through SUMO was input to the Wireless Host module via the Mobility sub-module. Then, the Wireless Host module conducted wireless communication, including MAC control and propagation characteristics of the wireless signal.

On the other hand, the ground station was constructed as a ground-fixed station module. The configuration of the ground fixed station module is shown in Figure 9. The ground fixed station module consists of a basic Mobility sub-module and a Wireless Host sub-module. Assuming that the ground station is fixed, the position information was passed to the Wireless Host sub-module, and wireless communication was simulated, similar to the LEO satellite module and the mobile station (vehicle) module.

IV. Simulation Evaluation

In this section, the settings for the simulation scenarios, and corresponding results are discussed. The common simulation parameters are shown in Table I. The simulation system is assumed to operate at center frequency 13.5 GHz Ku-band and 28.5 GHz Ka-band for service link and feeder link respectively. The system bandwidth is 250MHz for the service link, while the bandwidth is 500 MHz for the feeder link. Moreover, the LEO satellite system is assumed to follow a synthetic constellation, composed of 166 satellites that are distributed in 10 orbital planes inclined at 53°, with an orbital altitude around 550 km. The modulation schemes for the feeder link and service link are set as 256APSK and 16APSK respectively.

TABLE I
Simulation parameters

LEO satellite mobility module	
Orbit altitude	550 km
Apogee (farthest point from Earth)	547.9 km
Orbital inclination angle	53.2°
Orbital period	95.4 minutes
Communication related parameters	
Center frequency (GHz)	28.5
Bandwidth (MHz)	500
Ground station EIRP (dBW)	68.4
Feeder link Modulation	256APSK
Center frequency (GHz)	13.5
Bandwidth (MHz)	250
Satellite (Service link) EIRP (dBW)	36.7
Service link Modulation	16APSK
Path Loss Model	3GPP(TR38.811)
Traffic	UDP

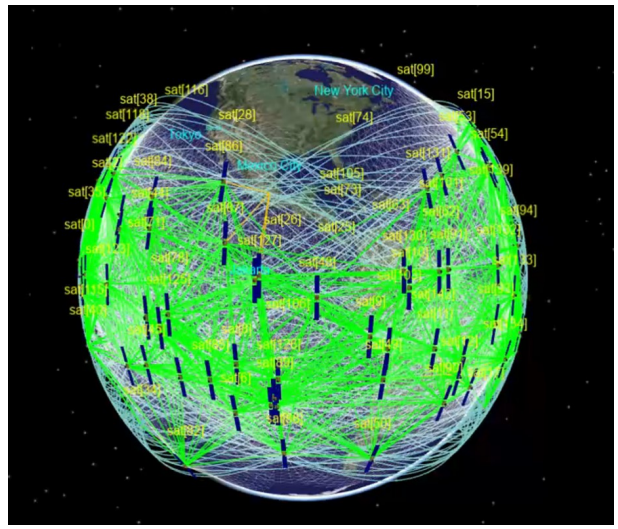

Fig. 10. Snapshot of the synthetic LEO satellite constellation

A. LEO satellite constellation mobility results

The simulation results of the LEO satellite constellation are shown in Figure 10, 11 and 12. The snapshot of the synthetic satellite constellation is shown in Figure 10 which is generated by using the OMNEST 3D visualization tool based on OpenSceneGraph and osgEarth libraries. Meanwhile, the footprint of a satellite is shown in Figure 11, where the satellite repeats the orbit plane in 14 cycles. While in the space above

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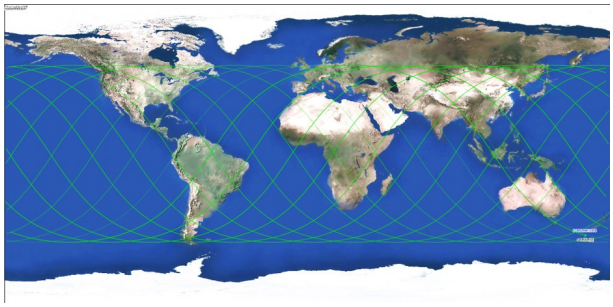

Fig. 11. Footprint of a LEO satellite

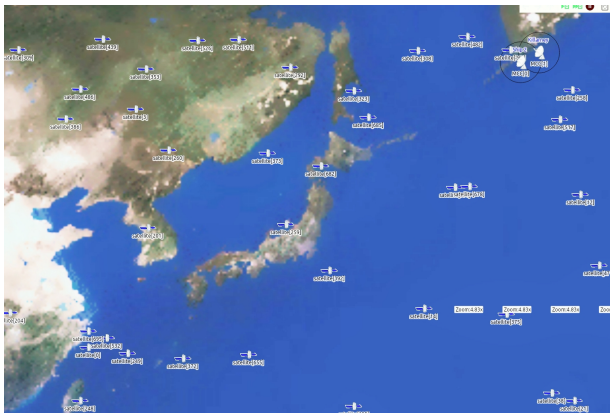

Fig. 12. Snapshot of the LEO satellites over Japan

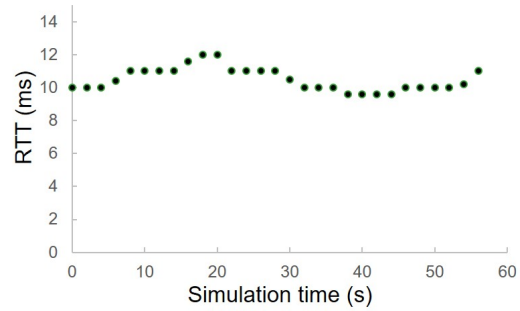

Fig. 13. The RTT performance

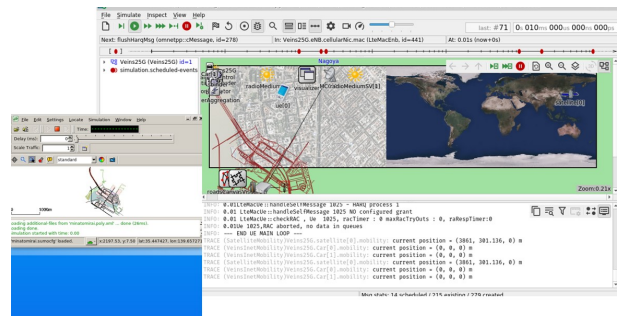

Fig. 14. The screenshot of the integrated simulator

and near Japan, the snapshot of the LEO satellites are shown in Figure 12.

Based on the synthetic LEO satellite constellation model, we first evaluated the Round Trip Time (RTT) with the PING application. The RTT is defined from the PING packet that was sent by the connected vehicle, responded by the ground station, and finally received by the connected vehicle. According to the results shown in Figure 13, the RTTs are around 10ms in the evaluation. In this sense, the propagation delay should be the major factor of the RTT in the evaluation. And the RTTs gradually increased in the first 20 seconds of simulation time, this is because the distance between the satellite and the ground station was getting further while the satellite was flying away. Then we can see from Figure 13 at a point, i.e. 20s simulation time shown in the results, the RTT suddenly decreased to a lower value because the ground station was handed over to another nearest satellite to keep the communication connection.

B. Satellite communication simulation results

Furthermore, we constructed several scenarios step-by-step to evaluate the LEO satellite communication with connected vehicles using the parameters shown in Table I which are based on the publicly available Starlink system. The demo simulation evaluation was conducted with UDP traffic was set. And the conducted initial evaluation results are discussed. The developed simulation platform screen is shown in Figure 14. The left side of Figure 14 shows the SUMO screen and the right side shows the screen displayed in OMNEST by mapping the coordinates from SUMO.

To evaluate the LEO satellite communication of the developed simulation platform, we first constructed demo scenario A in which one beam and one vehicle moving in the beam as shown in Figure 15. The vehicle is moving in a linear mobility way with a speed of 40km/h. The demo simulation evaluation was conducted with UDP traffic, and a packet size of 2187 bytes was set. The simulation results showed 218Mpbs throughput in average. In addition, for the delay, the main factor was radio propagation delay, which resulted in a delay of approximately 20ms between the vehicle and the satellite. To evaluate the implemented multi-beam function, we constructed two beams (multi-beam function) and one car in each beam (demo scenario B) as shown in Figure 16. The simulation parameters for demo scenario B can be found in Table I. According to the results shown in Figure 17, the average throughput for downlink is 228Mbps, while the average throughput for uplink was 98Mbps. In addition, for the delay, the main factor was radio propagation delay, which resulted in a delay of approximately 20ms. Based on the reference values of Starlink's practical measurements, the validity of the developed integrated simulator was demonstrated.

In addition, simulation evaluation was performed for a scenario (Demo Scenario C) where multiple users exist in a single beam. In Demo Scenario C, as shown in Figure 18, four users were set to exist in a single beam, and communication with the satellite was performed using the FDMA MAC control method.

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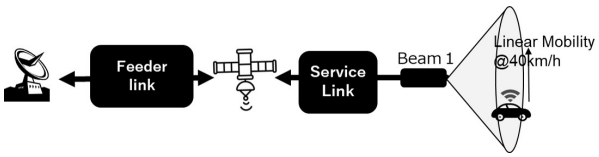

Fig. 15. Demo Scenario A

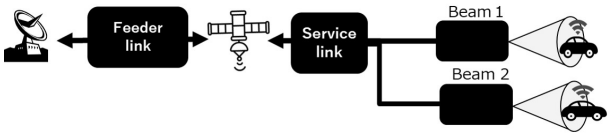

Fig. 16. Demo Scenario B

The evaluation results are presented in Figure 19. In Figure 19, the total upstream throughput was approximately 211Mbps and the total downstream throughput was approximately 270Mbps. Furthermore, upon examining the breakdown of downstream throughput, each user terminal (UT) achieved a throughput of approximately 67Mbps. These results indicate that the multi-access scheme effectively enabled each UT to communicate fairly with the satellite.

Conclusion

In this study, we proposed and developed an integrated simulator to evaluate LEO satellite communications for connected vehicles. We demonstrated the LEO satellite constellation and mobility using the developed simulator and further investigate the throughput performance of connected vehicles, which validated the functionality and effectiveness of the integrated simulator. Moving forward, we plan to add beam handover functions and conduct larger-scale network evaluations for connected vehicles.

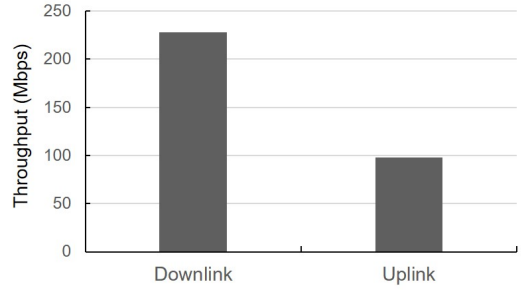

Fig. 17. The simulation results of demonstration scenario B

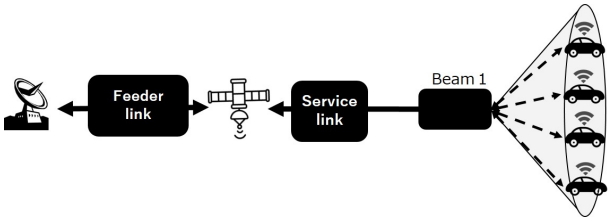

Fig. 18. Demonstration Scenario C

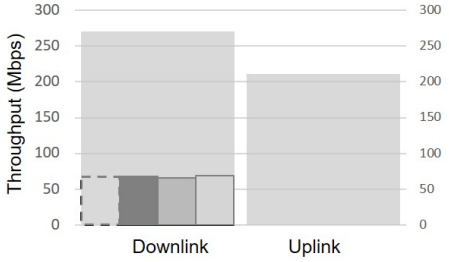

Fig. 19. The simulation results of demonstration scenario C

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